

Usable Without Touch? Revealing the Relationship Between Usability and Mid-Air Gestures

Proceedings of the Human Factors and Ergonomics Society Annual Meeting 2025, Vol. 69(1) 968–974
Copyright © 2025 Human Factors and Ergonomics Society
DOI: 10.1177/10711813251369780
journals.sagepub.com/home/pro



Justin Catalano¹  and Yue Luo¹ 

Abstract

Mid-air gestures are an interaction method where the participant uses their hand(s) to perform a movement or sequence that represents control. This method of interaction is becoming more common for public use, yet the impact on usability lacks summarization. This systematic review explores themes of usability testing methods, applications, gesture design, multimodal combinations, input comparisons, and the participants. Findings reveal that the usability outcomes (effectiveness, efficiency, and user satisfaction) vary based on the type of evaluation method, application, design of gestures, and testing methods used. Furthermore, participant differences such as prior experience, training, and physical ability affect perceived usability. These insights reveal the need for inclusive, adaptive, and context-aware design strategies. Future research should standardize usability evaluation practices and prioritize user-centered approaches to mid-air gesture development. We propose guidelines for future research that will improve gesture systems, feedback, contextual fit, and usability evaluations.

Keywords

mid-air gestures, usability, human-computer interaction

Introduction

Mid-air gestures are an emerging modality of human-computer interaction (HCI) that allows for a touch-free experience with a system (Koutsabasis & Vogiatzidakis, 2019). These gestures (i.e., pointing, pinching, and swiping in mid-air) are used to create a more natural and intuitive user experience. Current examples of mid-air gestures in the public domain are: AppleWatch's Assistive Touch feature, Apple Vision Pro, BMW iDrive, and Samsung TVs. Despite growing interest and implementation, the usability of mid-air gesture interaction remains underexplored. Much of the existing literature focuses on technical aspects such as gesture recognition accuracy or algorithm development, often overlooking how users perceive and experience this interaction (Chen et al. 2017). This results in a limited understanding of key dimensions in usability, which can be defined as effectiveness, efficiency, and user satisfaction (Bevan et al., 1991).

Our goal is to assess if the participants perceive mid-air gestures as usable based on their subjective experience, rather than relying on performance-based indicators of usability. To do this, we seek to investigate the relationship between usability and mid-air gestures through; (1) Type of Research: How and where usability is addressed, (2) Gesture System Design: What approaches and how complex was the mid-air gesture system, and (3) Participant Factors: How usability

varied based on discovered differences. In doing so, we will address the following research questions: RQ1: What has been discovered about usability of mid-air gesture interactions? RQ2: What are common patterns and gaps that future mid-air gesture research addresses?

Method

This study follows the Preferred Report of Items for Systematic Reviews and Meta-Analyses (PRISMA) (Page, 2021) protocol to gather studies. We used the databases of PsycINFO, Engineering Village (Compendex), Web of Science, Science Direct, and PubMed in October 2024. The search string used across these databases were: (“Mid-air gesture*” OR “air gesture”) AND (“Usability” OR “Ease of Use” OR “User Satisfaction” OR “Efficiency” OR “Error Prevention” OR “Satisfaction” OR “Learnability”) using the Title/Abstract, previous 10 years (2013–2024), and English only filters. A total of 218 studies were retrieved. After

¹San José State University, San José, CA, USA

Corresponding Author:

Justin Catalano, Department of Industrial and Systems Engineering, San José State University, 1 Washington Sq, San José, CA 95192-0080, USA.
Email: justin.catalano@sjsu.edu

Table 1. Inclusion/Exclusion Criteria.

Item	Results
I1.	Journal, Article or Proceeding
I2.	Include an experimental design of $n < 1$
I3.	Must be quantitative, qualitative, or mixed methods results
I4.	Usability is defined as effectiveness, efficiency, and user satisfaction
I5.	Has an application domain
I6.	Can paired with other sensory modalities
I7.	Can be paired with other interaction methods
E1.	Referring to their own product or service to remove bias

removing duplicates, we systematically screened the studies based on the inclusion/exclusion criteria viewable in Table 1, with “I” representing inclusion and “E” representing exclusion.

After completing the screening, the first author identified themes and pulled metadata to categorize usability findings based on gesture system design, types of research, and participant factors. The PRISMA result is viewable in Figure 1.

Results

Ten studies, viewable in Table 2, were selected for this systematic review and analyzed for the themes and subthemes of gesture system design, types of research, and participant factors.

Research Methods

Usability Evaluations. All 10 studies included a usability test, with the breakdown viewable in Table 3. These test include: The System Usability Scale (SUS) ($n=4$), the Ease of Use scale (EoU) ($n=1$), the System Usability Scale/Gesture Usability Scale (SUS/GUS) ($n=1$), the Short User Experience Questionnaire (UEQ-S) ($n=1$), or a self-designed subjective usability test (SDSUT), commonly evaluating comfort, performance, and user preference gestures usability ($n=3$) (Cui et al., 2023; Vaidyanathan & Rosenberg, 2014; Zhang et al., 2015). Although there was variation in the type of evaluation, all the studies captured subjective perceptions of usability from participants.

Applications. The 10 studies reviewed reveal four applications evaluating the impact of mid-air gestures on usability: Automobiles have subgroups of autonomous driving ($n=2$) and in-vehicle navigation systems (IVNS, $n=3$), Public Displays ($n=2$), Remote Control ($n=2$), and Virtual Environments ($n=1$). Usability results for are viewable in Table 3.

Gesture System Design

Elicitation Studies. The mid-air gestures were designed through three primary approaches: User-elicited ($n=2$), author-defined ($n=4$), literature-based ($n=4$).

User Elicited. Elicitation studies asked users to generate mid-air gestures based on the intended interaction. Guérit et al. (2019) had participants ($n=19$) generate 57 waving and directional gestures, resulting in an average usability (SUS). Fink et al. (2023) had participants ($n=15$) generate 14 single-finger gestures that were found easier to use than compounded gestures, but gestures overall were found easy to use (EoU).

Author-defined. In these studies, the authors created the mid-air gestures. They created mid-air gestures with fingers (Vaidyanathan & Rosenberg, 2014; Waugh and Robertson, 2021) and full hand (Zhang et al., 2015). The studies found that gestures were not the preferred interaction method (Vaidyanathan & Rosenberg, 2014; Zhang et al., 2015), with mixed usability results (SUS/GUS ($n=1$), SDSUT ($n=2$)). Kronester et al. (2021) used grabbing gestures, resulting in a usable outcome (UEQ-S).

Literature-based. Cui et al. (2023) refined 30 gesture sets from prior studies, selecting 12 maneuver-based finger gestures with high usability. Cao et al. (2024), Detjen et al. (2020), and Tabbarah et al. (2023) adapted full-hand gestures from previous research, with SUS scores ranging from 68 to 81 (mean unprovided).

Modalities Paired with Gestures. Five studies reveal three modalities paired with mid-air gestures: Audio, Haptic, and Gaze.

Audio. Audio feedback improved mid-air gesture usability across studies. Cao et al. (2024) found acceptable usability with a heads-up display (SUS, $m=68.85$). Tabbarah et al. (2023) reported higher SUS scores with spearcons in generative interfaces ($m=81$ adaptive, $m=76$ fixed) than other audio feedback. Fink et al. (2023) found high ease of use (EoU $m=5$) with audio over other modalities.

Haptic. Fink et al. (2023) found that adding haptics did not improve ease of use (EoU $m=3.5$) compared to only mid-air gesture-audio control, but they did not compare a baseline non-haptic mid-air gesture interaction. In contrast, Kronester

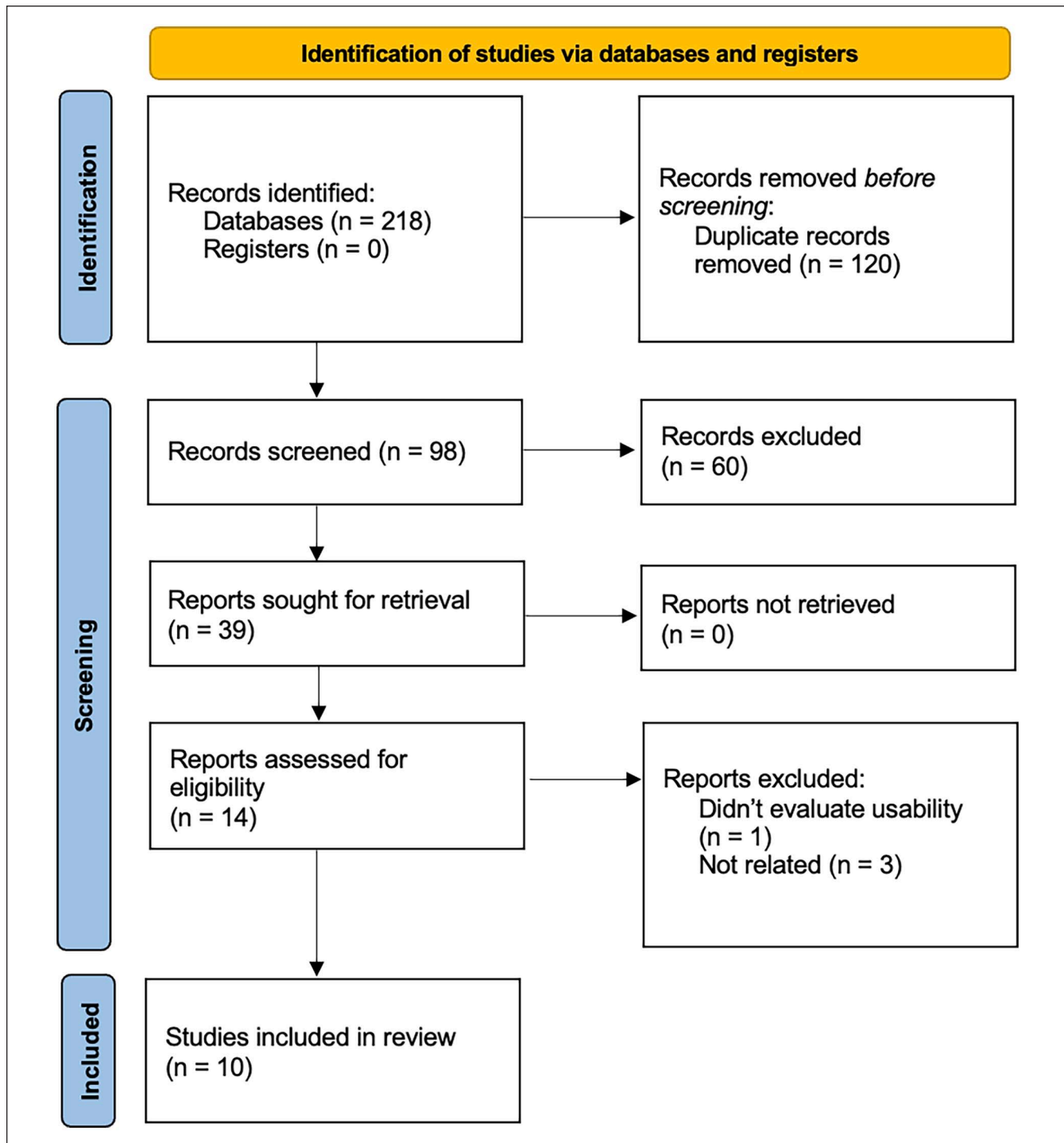


Figure 1. PRISMA search.

et al. (2021) found that the inclusion of haptic feedback on the wrist had a significantly better experience than no haptic feedback (UEQ-S).

Gaze. Zhang et al. (2015) found that combining gaze with mid-air gestures was preferred over mid-air gestures alone (SDSUT).

Modalities Compared. Four studies compared mid-air gestures to three interaction methods: Touch, Voice, and Gaze-hand.

Touch. Three studies found that participants preferred touch over mid-air gestures with mixed usability results. Detjen et al. (2020) reported similar SUS scores for touch and mid-air gestures ($m \leq 70$), with touch favored. Vaidyanathan and

Table 2. Studies Selected.

Item	Results
1.	Cao et al. (2024)
2.	Cui et al. (2023)
3.	Detjen et al. (2020)
4.	Fink et al. (2023)
5.	Guérit et al. (2019)
6.	Kronester et al. (2021)
7.	Tabbarah et al. (2023)
8.	Vaidyanathan and Rosenberg (2014)
9.	Waugh and Robertson (2021)
10.	Zhang et al. (2015)

Rosenberg (2014) found that 90% ($n=30$) preferred touch interactions. Waugh and Robertson (2021) reported higher SUS/GUS scores for touch ($m=90/m=95$) over mid-air gestures.

Voice. Detjen et al. (2020) found that voice interaction had the same usability as mid-air gestures (SUS, $m < 70$) and held the same preference as touch over mid-air gestures.

Gaze-hand. Zhang et al. (2015) found that including gaze in mid-air gesture interaction was better (SDSUT, 86%, $n=15$) than just hand-only.

Participant Factors

A total of 223 participants evaluated the usability of mid-air gestures in the 10 studies with varying factors.

Participant Differences. Four studies evaluated the difference between user characteristics. Right vs. left-handedness (Cui et al., 2023; Waugh & Robertson, 2021), international and cultural identity (Tabbarah et al., 2023), or blind and visually impaired (BVI) (Fink et al., 2023), which did not impact usability scores (measured via SDSUT, SUS/GUS, and EoU, respectively).

Familiarity. Three studies evaluated prior experience with mid-air gestures. Detjen et al. (2020) and Kronester et al. (2021) reported a wide range of familiarity with overall positive usability ratings (UEQ-S). Vaidyanathan & Rosenberg (2014) found that 69% of participants ($n=32$) were unfamiliar and preferred touch (SDSUT).

Training. Four studies included practice sessions or learnability tasks to improve user competency (Cao et al., 2024; Fink et al., 2023; Vaidyanathan & Rosenberg, 2014; Zhang et al., 2015).

Discussion

Context of Use

We discovered that there were various usability evaluations used to evaluate the usability of mid-air gestures, leading to

an inconclusive “global” view of usability. The complexity of mid-air gesture interaction may exceed what current usability tests can reveal, which is why using a standardized SUS or SUS/GUS might not be effective (Cao et al., 2024; Detjen et al., 2020; Guérit et al., 2019; Tabbarah et al., 2023; Waugh & Robertson, 2021). (1) Future research should develop a usability framework that captures the complexities of mid-air gesture interaction, which could include utility, learnability, understandability, satisfaction, efficiency, comfort, and memory (Tabbarah et al., 2023; Vaidyanathan & Rosenberg, 2014; Waugh & Robertson, 2021; Zhang et al., 2015) (Table 4). Usability varied by application when mid-air gestures were the primary interaction method. Automobiles showed a promising domain due to a reduction of visual or auditory attention, but that could be countered by a lack of spatial awareness (Cao et al., 2024; Cui et al., 2023; Detjen et al., 2020; Fink et al., 2023). Public displays could benefit from hygienic use. However, when systems aren't inherently appropriate for mid-air interaction, a reduction of usability could occur (Vaidyanathan & Rosenberg, 2014; Waugh & Robertson, 2021). Remote controls can benefit from the increased range and customized interaction, which could pose issues in performance and confirmation (Guérit et al., 2019; Zhang et al., 2015). Virtual environments are a promising application and can benefit from additional feedback when interacting with a virtual object. (Kronester et al., 2021). Future research should (2) weigh the benefits and challenges of mid-air gestures within an application (Table 4).

Design of Mid-Air Gesture Systems

The type of movement, gesture creation method, and feedback are factors that contribute to the usability of mid-air gestures. Finger gestures allow for precise interaction at a distance, with higher usability in smaller interfaces (Cui et al. 2023) than larger interfaces (Vaidyanathan & Rosenberg, 2014; Waugh & Robertson, 2021). The method of gesture design was not a contributing factor. Most full-hand gestures were literature-based and showed acceptable usability, common in the automobile context with average usability in small screens (INVS: Cao et al., 2024; Tabbarah et al., 2023. Autonomous Driving: Detjen et al., 2020), with some usability limitations in ranged interaction (Zhang et al., 2015). Whole-arm gestures worked well in confined spaces (EoS: Fink et al., 2023). Future researchers should (3) use application specific mid-air gestures (Table 4). Results suggest that pairing mid-air gestures with additional modalities can significantly improve usability. Audio feedback, especially spearcons, enhanced the understanding of the change of systems status, but these audio components should be complementary to the environment and goals of the participant (Cao et al., 2024; Tabbarah et al., 2023; Fink et al., 2023). Haptic feedback also helped the usability of mid-air gestures by addressing the sensory gap in mid-air gesture interactions (Fink et al., 2023; Kronester et al., 2021). Similarly, when gaze and mid-air gestures are combined, it improves spatial

Table 3. Usability and Application Results Grouped by Usability Evaluation.

Test		Results		
SUS (<i>n</i> = 4)	Autonomous driving Average usability (<i>m</i> ≤ 70) Detjen et al. (2020)	IVNS Acceptable usability (<i>m</i> = 68.85) Cao et al. (2024)	IVNS Acceptable usability (68-81) Tabbarah et al. (2023)	Remote control Average usability (<i>m</i> = 70.92, <i>n</i> = 11) Guérit et al. (2019)
SDSUT (<i>n</i> = 3)	IVNS High subjective usability (<i>m</i> = 89 > 75) Cui et al. (2023)	Public displays Less subjective usability (90% agree, <i>n</i> = 30) Vaidyanathan and Rosenberg (2014)	Remote Control Usable, not preferred (<i>n</i> = 13) Zhang et al. (2015)	
EoU (<i>n</i> = 1)	Autonomous driving High EoS (<i>m</i> = 5) Fink et al. (2023)			
SUS/ GUS (<i>n</i> = 1)	Public displays Mixed usability (SUS <i>m</i> = 70, GUS <i>m</i> = 64) Waugh and Robertson (2021)			
UEQ-S (<i>n</i> = 1)	Virtual environments Usable (<i>n</i> = 144) Kronester et al. (2021)			

Note. SUS = System Usability Scale; GUS = Gesture Usability Scale; EoU = Ease of Use; SDSUT = Self-Designed Subjective Usability Test; UEQ-S = User Experience Questionnaire (Short).

Table 4. Guidelines for Future Research.

Item	Guidelines for future research
1.	Include usability metrics that cover utility, learnability, understandability, satisfaction, efficiency, comfort, and memory
2.	Assess benefits and challenges of adding mid-air gesture interaction
3.	Use mid-air gestures that are application specific
4.	Pair at least one feedback modality to enhance mid-air gesture comprehension.
5.	Evaluate the environmental factors that may affect the usability of mid-air gestures.
6.	Insure an inclusive interaction
7.	Test and reduce the effects of familiarity
8.	Include training to increase comprehension

awareness and usability in precision-focused tasks (Zhang et al., 2015). Multimodal approaches further enhanced usability, showing that combining multiple feedback types can optimize the overall interaction experience (Cao et al., 2024; Cui et al., 2023; Fink et al., 2023). Future research should (4) pair at least one feedback modality to enhance mid-air gesture comprehension (Table 4). However, when compared to other interaction methods (touch, voice, gaze), mid-air gestures were generally less preferred. Touch had the highest preference because it is a more common and intuitive interaction method than mid-air gestures (Detjen et al., 2020; Vaidyanathan & Rosenberg, 2014; Waugh & Robertson, 2021), but mid-air gestures may prove more viable in contexts like driving (Detjen et al., 2020). Voice input was preferred but faced challenges in noisy environments, unlike mid-air gestures (Detjen et al., 2020). Gaze-hand interactions were preferred over mid-air gestures alone, likely due to

gaze providing direct visual feedback that alleviates spatial awareness issues (Zhang et al. 2015). Future research should (5) evaluate the environmental factors that may affect the usability of mid-air gestures (Table 4).

Participant Factors

Participant factors like individual differences, familiarity, and training influenced mid-air gesture usability, but without consistent patterns. Studies that evaluated participants' differences, such as right vs left-handedness or BVI, had generally positive usability outcomes. Regardless of usability outcome, future research should (6) ensure that the mid-air gesture is inclusive (Cui et al., 2023; Tabbarah et al., 2023; Fink et al., 2023) (Table 4). Familiarity with mid-air gestures didn't necessarily improve usability, as users familiar with traditional interfaces rated mid-air gestures similarly or lower (Detjen

et al., 2020; Kronester et al., 2021). Future research should aim to (7) test and reduce the effects of familiarity to reveal how feasible further public-facing integration will be (Table 4). Training also showed mixed results, improving usability in some contexts but not others (Cao et al., 2024; Fink et al., 2023; Vaidyanathan & Rosenberg, 2014). Future research should (8) include training to help with the comprehension of mid-air gesture interaction (Table 4).

Proposal of Guidelines

Based on our discoveries, we propose eight guidelines for future research based on the results and discussion of this systematic review, which is viewable in Table 4. These guidelines can help researchers adopt a more comprehensive and nuanced approach to studying and designing mid-air gesture interactions.

Conclusions

This review focuses on the participants perceived usability of systems that use mid-air gestures to interact with computers. We highlight the complexity of evaluating and designing mid-air gesture interactions. Usability varies significantly based on gesture design, application type, evaluation methods, and user differences such as experience, training, and ability. While mid-air gestures offer hands-free interaction potential, they often fall short compared to more familiar modalities like touch or voice. To enhance usability, future systems must prioritize inclusive, adaptive, and context-aware designs. Standardizing usability testing and incorporating user feedback into gesture creation will help bridge gaps in intuitiveness and learnability. By addressing these challenges, mid-air gestures can become more practical, accessible, and user-centered across diverse applications.

Limitations

This review only includes English papers from 2013 to 2024, which could leave out informational studies published in different years and languages. An analysis of performance was also purposefully left out of this review based on the scope and insights sought out to achieve, but the authors recognize the important relationship that usability and performance have. Future research could evaluate the overall relationship between usability and performance to gain further insight into mid-air gestures as an interaction method. The authors also didn't rate the quality of papers which should be taken into consideration.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

ORCID iDs

Justin Catalano  <https://orcid.org/0009-0001-4936-9859>

Yue Luo  <https://orcid.org/0000-0002-0062-3190>

References

- Bevan, N., Kirakowski, J., & Maisse, L. J. (1991). What is usability. In *Proceedings of the 4th International Conference on Human-Computer Interaction* (HCI International '91, Stuttgart, Germany). Taylor & Francis. <https://doi.org/10.1201/9781482272574-76>
- Cao, Y., Li, L., Yuan, J., & Jeon, M. (2024). Head-up displays improve drivers' performance and subjective perceptions with the in-vehicle gesture interaction system. *International Journal of Human-Computer Interaction*, 1–15. <https://doi.org/10.1080/10447318.2024.2387397>
- Chen, L.-C., Cheng, Y.-M., Chu, P.-Y., & Sandnes, F. E. (2017). Identifying the usability factors of mid-air hand gestures for 3D virtual model manipulation. In M. Antona & C. Stephanidis (Eds.), *Universal access in human-computer interaction. Designing novel interactions* (UAHCI 2017, Lecture Notes in Computer Science, Vol. 10278, pp. 393–402). Springer. https://doi.org/10.1007/978-3-319-58703-5_29
- Cui, C., Shen, G., Wang, Y., Xu, Y., Du, H., Zhang, W., & Kong, X. (2023). Impact of in-air gestures on in-car task's driver distraction. *Electronics*, 12(7), 1626. <https://doi.org/10.3390/electronics12071626>
- Detjen, H., Geisler, S., & Schneegass, S. (2020). Maneuver-based control interventions during automated driving: Comparing touch, voice, and mid-air gestures as input modalities. In *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* (pp. 3268–3274). IEEE. <https://doi.org/10.1109/SMC42975.2020.9283431>
- Evan, N., Kirakowski, J., & Maisse, L. J. (1991). What is usability. In *Proceedings of the 4th International Conference on Human-Computer Interaction* (HCI International '91, Stuttgart, 1991). Taylor & Francis. <https://doi.org/10.1201/9781482272574-76>
- Fink, P., Dimitrov, V., Yasuda, H., Chen, T. L., Corey, R. R., Giudice, N. A., & Sumner, E. S. (2023). Autonomous is not enough: Designing multisensory mid-air gestures for vehicle interactions among people with visual impairments. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI '23, Article 74, pp. 1–13). ACM. <https://doi.org/10.1145/3544548.3580762>
- Guérit, R., Cierro, A., Vanderdonckt, J., & Pérez-Medina, J. L. (2019). Gesture elicitation and usability testing for an armband interacting with Netflix and Spotify. In *Advances in Intelligent Systems and Computing* (Vol. 918, pp. 625–637). Springer. https://doi.org/10.1007/978-3-030-11890-7_60
- Koutsabasis, P., & Vogiatzidakis, P. (2019). Empirical research in mid-air interaction: A systematic review. *International Journal of Human-Computer Interaction*, 35(18), 1747–1768. <https://doi.org/10.1080/10447318.2019.1572352>
- Kronester, M. J., Riener, A., & Babic, T. (2021). Potential of wrist-worn vibrotactile feedback to enhance the percep-

- tion of virtual objects during mid-air gestures. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Article 256, pp. 1–7). ACM. <https://doi.org/10.1145/3411763.3451655>
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., & Mulrow, C. D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, 372, n71. <https://doi.org/10.1136/bmj.n71>
- Tabbarah, M., Cao, Y., Shamat, A. A., Fang, Z., Li, L., & Jeon, M. (2023). Novel in-vehicle gesture interactions: Design and evaluation of auditory displays and menu generation interfaces. In *Proceedings of the 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (AutomotiveUI '23, pp. 224–233). ACM. <https://doi.org/10.1145/3580585.3607164>
- Vaidyanathan, V., & Rosenberg, D. (2014). “Will use it, because I want to look cool”: A comparative study of simple computer interactions using touchscreen and in-air hand gestures. In *Human–Computer Interaction. Advanced interaction modalities and techniques* (HCI 2014, Lecture Notes in Computer Science, Vol. 8511, pp. 170–181). Springer. https://doi.org/10.1007/978-3-319-07230-2_17
- Waugh, K., & Robertson, J. (2021). Don’t touch me! A comparison of usability on touch and non-touch inputs. In C. Ardito, et al. (Eds.), *Human–Computer Interaction – INTERACT 2021: 18th IFIP TC 13 International Conference* (Lecture Notes in Computer Science, Vol. 12936, pp. 400–404). Springer. https://doi.org/10.1007/978-3-030-85607-6_46
- Zhang, Y., Stellmach, S., Sellen, A., & Blake, A. (2015). The costs and benefits of combining gaze and hand gestures for remote interaction. In *Human–Computer Interaction – INTERACT 2015* (Lecture Notes in Computer Science, Vol. 9298, pp. 570–577). Springer. https://doi.org/10.1007/978-3-319-22698-9_39